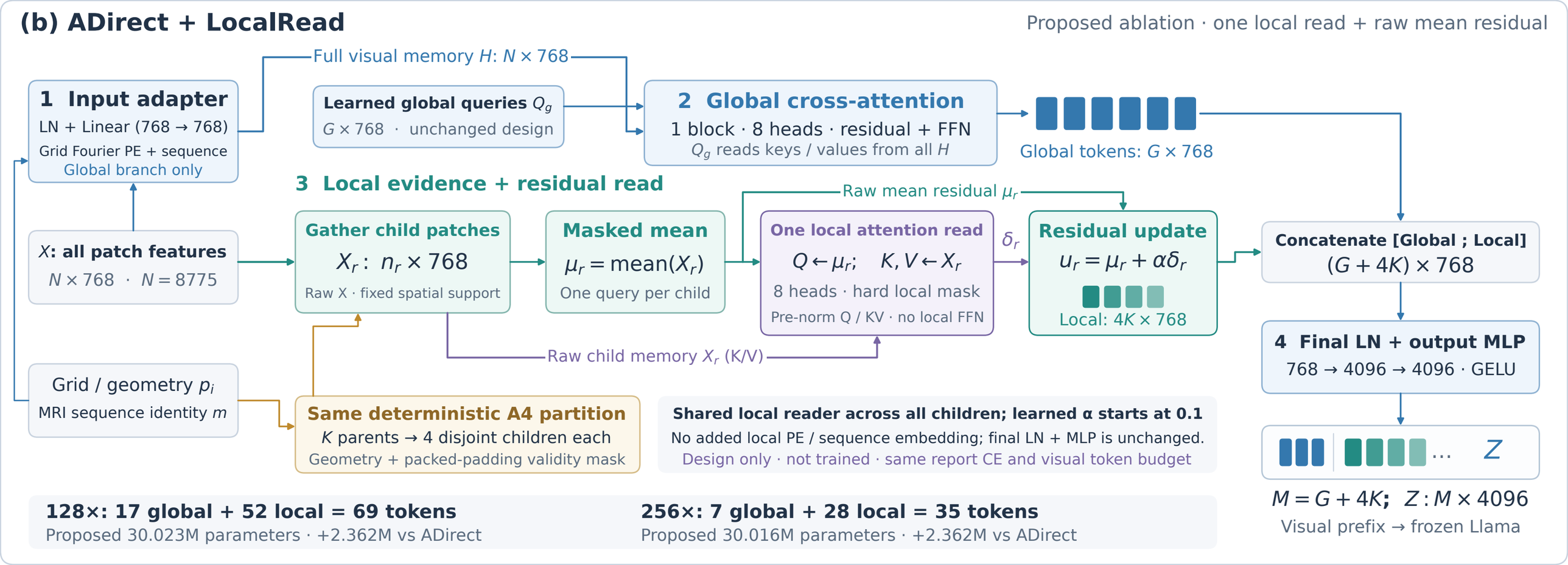
(b) ADirect + LocalRead
Proposed ablation · one local read + raw mean residual
Full visual memory H: N × 768
1 Input adapter
LN + Linear (768 → 768)
Grid Fourier PE + sequence
Global branch only
Learned global queries Q_g
G × 768 · unchanged design
2 Global cross-attention
1 block · 8 heads · residual + FFN
Q_g reads keys / values from all H
Global tokens: G × 768
3 Local evidence + residual read
Raw mean residual μ_r
X: all patch features
N × 768 · N = 8775
Gather child patches
X_r : $n_r \times 768$
Raw X · fixed spatial support
Masked mean
$\mu_r = \mathrm{mean}(X_r)$
One query per child
One local attention read
$Q \leftarrow \mu_r$; $K, V \leftarrow X_r$
8 heads · hard local mask
Pre-norm Q / KV · no local FFN
δ_r
Residual update
$u_r = \mu_r + \alpha\delta_r$
Local: 4K × 768
Raw child memory X_r (K/V)
Concatenate [Global ; Local]
$(G + 4K) \times 768$
4 Final LN + output MLP
768 → 4096 → 4096 · GELU
Grid / geometry p_i
MRI sequence identity m
Same deterministic A4 partition
K parents → 4 disjoint children each
Geometry + packed-padding validity mask
Shared local reader across all children; learned α starts at 0.1
No added local PE / sequence embedding; final LN + MLP is unchanged.
Design only · not trained · same report CE and visual token budget
Z
$M = G + 4K$; $Z : M \times 4096$
Visual prefix → frozen Llama
128×: 17 global + 52 local = 69 tokens
Proposed 30.023M parameters · +2.362M vs ADirect
256×: 7 global + 28 local = 35 tokens
Proposed 30.016M parameters · +2.362M vs ADirect